

Current Progress

- Implemented **real-time scanning** of active webpages using DOM analysis and URL heuristics.
- Developed **popup interface** with functionality to:
 - Scan the current page
 - Analyze manually pasted URLs
- Added **basic detection rules**, including:
 - Suspicious keywords in URLs
 - Presence of misleading domain structures (e.g., excessive subdomains)
 - Detection of insecure protocols (HTTP vs HTTPS)
- Integrated **warning system**, including visual flags and risk indicators in the UI.
- Began implementing **local storage of recent scans** for user reference.

Challenges Encountered

- **False positives**: Legitimate sites (e.g., email or cloud storage platforms) are occasionally flagged as suspicious due to strict heuristic rules.
- **Performance considerations**: Continuous real-time scanning introduces potential overhead on complex or resource-heavy websites.
- **DOM variability**: Different website structures make it difficult to create universally reliable detection patterns.

Solutions & Mitigations

- Refining detection thresholds to reduce false positives while maintaining sensitivity.
- Planning to introduce a **whitelist system** for trusted domains.
- Optimizing scanning logic to minimize performance impact (e.g., limiting scan frequency or scope).
- Expanding rule set to include more context-aware checks rather than simple keyword matching.

Next Steps

- Improve **accuracy of detection algorithm** using weighted scoring instead of binary flags.
- Finalize **recent scan history feature** with persistent storage.
- Implement **accessibility improvements**, including clearer visual indicators and color-safe UI design.
- Conduct **testing on known vulnerable sites** (e.g., OWASP Juice Shop) to validate detection effectiveness.
- Prepare **demo workflow** showcasing real-time detection and user interaction.

Overall Status

The project is on track, with core functionality implemented and actively being refined. The focus is shifting from feature development to optimization, accuracy, and user experience improvements ahead of final presentation and evaluation.

